

JESS Thermodynamic Database v8.9

Iridium

24-Jan-23

Reaction No. 5074							
$\text{Ir+3_Cl-1(6)} + 3\text{e-1} = \text{Ir(s)} + 6\text{Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 39. (0.5SD)$	3	ENS	802
2	25	0	Inf. Dilution	$E: 0.77 (2SF)$	3	CRV	6330
3	25	0	Inf. Dilution	$E: 0.83 (2SF)$	0	CRV	22551

Reaction No. 5075							
$\text{Ir+3_Cl-1(6)} = \text{Ir+3_Cl-1(5)} + \text{Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1.03	HClO ₄	$\lg K: 1.08 (0.05SD)$	5	MKN	874
2	25	3	HClO ₄	$\lg K: 0.8 (0.05SD)$	4	EIE	899
3	35	1	HClO ₄	$\lg K: 1.4 (0.05SD)$	5	MKN	875
4	35	3	HClO ₄	$\lg K: 0.7 (0.05SD)$	5	MKN	875
5	42	1	HClO ₄	$\lg K: 1.3 (0.05SD)$	5	MKN	875
6	42	3	HClO ₄	$\lg K: 0.7 (0.05SD)$	5	MKN	875
7	50	1	HClO ₄	$\lg K: 1.2 (0.05SD)$	5	MKN	875
8	50	2.2	NaClO ₄	$\lg K: 0.9 (0.05SD)$	5	MSP	873
9	50	3	HClO ₄	$\lg K: 0.6 (0.05SD)$	5	MKN	875
10	50	3.7	NaClO ₄	$\lg K: 0.4 (0.05SD)$	5	MSP	873
11	60	1	HClO ₄	$\lg K: 1.1 (0.05SD)$	5	MKN	875
12	60	3	HClO ₄	$\lg K: 0.6 (0.05SD)$	5	MKN	875

Reaction No. 5076							
$\text{Ir+3_Cl-1(4)} + \text{Cl-1} = \text{Ir+3_Cl-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	40:50	3.7	HClO ₄	$\lg K: 0.67 (2SF)$	5	MKN	872
2	40:50	3.7	NaClO ₄	$\lg K: 0.7 (0.05SD)$	5	MKN	872
3	50	2.2	HClO ₄	$\lg K: 0.55 (2SF)$	5	MKN	872
4	50	2.2	NaClO ₄	$\lg K: 0.6 (0.05SD)$	5	MKN	872

Reaction No. 5078							
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IrO2(s) + 4<e-1> + 4<H+1> = Ir(s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:50.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:62.6(0.05SD)	0	MDG	600
3	25	0	Inf. Dilution	E:0.935(0.005SD)	0	CRV	22551
Note (3): Given as a conditional E0' but without conditions							

Reaction No. 5079							
Ir+4_Cl-1(6) + e-1 = Ir+3_Cl-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.65(0.05SD)	5	MEF	878
2	25	0.1	NaCl	lgK:15.2(0.05SD)	5	MEF	879
3	25	1	HCl	lgK:15.8(0.05SD)	5	MEF	877
4	25	1	NaCl	lgK:15.8(0.05SD)	5	MEF	877
5	25	0	Inf. Dilution	E:0.8665(4SF)	4	CRV	6330
6	25	0.0042	HCl	E:0.882(3SF)	5	MEF	878
7	25	0.0084	HCl	E:0.887(3SF)	5	MEF	878
8	25	0.0223	HCl	E:0.893(3SF)	5	MEF	878
9	25	0.038	HCl	E:0.897(3SF)	5	MEF	878
10	25	0.058	HCl	E:0.902(3SF)	5	MEF	878
11	25	0.11	HCl	E:0.911(3SF)	5	MEF	878
12	25	0.3	HCl	E:0.867(3SF)	0	CRV	22551

Reaction No. 5080							
Ir+4 + e-1 = Ir+3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:21.5(0.05SD)	5	MPL	879

Reaction No. 5081							
Ir+4_Cl-1(6) + OH-1 = Ir+4_Cl-1(5)_OH-1 + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:7.08(0.1SD)	5	MKN	5026

Reaction No. 5082							
Ir+4_Cl-1(5)_OH-1 + OH-1 = Ir+4_Cl-1(4)_OH-1(2) + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:6.45(0.1SD)	5	MKN	5026

Reaction No. 5083							
$\text{Ir+4_Cl-1(4)_OH-1(2)} + \text{OH-1} = \text{Ir+4_Cl-1(3)_OH-1(3)} + \text{Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:6.04(0.1SD)	5	MKN	5026

Reaction No. 5084							
$\text{Ir+4_Cl-1(3)_OH-1(3)} + \text{OH-1} = \text{Ir+4_Cl-1(2)_OH-1(4)} + \text{Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:5.18(0.1SD)	5	MKN	5026

Reaction No. 5106							
$\text{Ru+4_Cl-1(5)_OH-1} + \text{Ir+4_Cl-1(5)} = \text{Ir+4_Ru+4_Cl-1(10)_OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	3	MCO	850

Reaction No. 5107							
$\text{Ru+4_Cl-1(4)_OH-1(2)} + \text{Ir+4_Cl-1(5)} = \text{Ir+4_Ru+4_Cl-1(9)_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	3	MCO	850

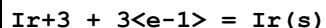
Reaction No. 5111							
$\text{Ru+4_Cl-1(3)_OH-1(3)} + \text{Ir+4_Cl-1(5)} = \text{Ir+4_Ru+4_Cl-1(8)_OH-1(3)}$							
Note: Weighted out coz its thermodynamically isolated							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	0	MCO	850

Reaction No. 5112							
$\text{Ru+4_Cl-1(2)_OH-1(4)} + \text{Ir+4_Cl-1(5)} = \text{Ir+4_Ru+4_Cl-1(7)_OH-1(4)}$							
Note: Weighted out coz its thermodynamically isolated							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	0	MCO	850

Reaction No. 5113							
$\text{Ru+4_Cl-1_OH-1(5)} + \text{Ir+4_Cl-1(5)} = \text{Ir+4_Ru+4_Cl-1(6)_OH-1(5)}$							
Note: Weighted out coz its thermodynamically isolated							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

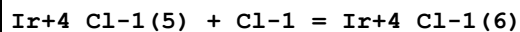
1	25	0	Inf. Dilution	lgK:5.9(0.2SD)	0	MCO	850
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Reaction No. 5118



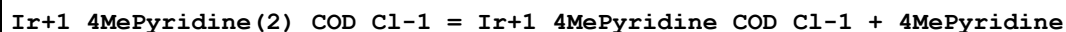
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:58.6(0.05SD)	4	ENS	802
2	25	0	Inf. Dilution	dG:-80.0(0.1SD)	5	ENS	5253
3	25	0	Inf. Dilution	E:1.156(4SF)	5	CRV	6330

Reaction No. 5119



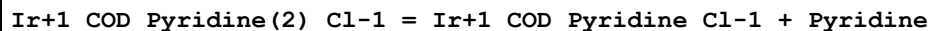
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.5(1SF)	1	ELC	
2	25	1	Unknown	lgK:-3.(0.8SD)	2	ESR	899

Reaction No. 38052



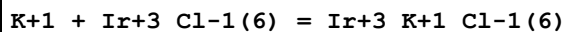
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Methanol	lgK:1.76(3SF)	5	MSP	583
2	25		Methanol	lgK:1.510(4SF)	5	MLC	583

Reaction No. 38053



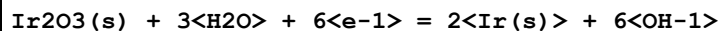
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Methanol	lgK:1.27(3SF)	5	MSP	583
2	25		Methanol	lgK:1.243(4SF)	5	MLC	583

Reaction No. 54701



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	42	3	LiCl	lgK:-0.3(0.05SD)	5	MPT	5025
2	50	3	LiCl	lgK:-0.1(0.05SD)	5	MPT	5025
3	42:50	3	LiCl	dH:13.(0.8SD)	4	EVH	899

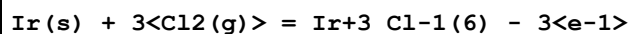
Reaction No. 54703



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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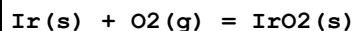
1	25	0	Inf. Dilution	lgK:9.9(0.1SD)	4	ENS	802
2	25	0	Inf. Dilution	E:0.098(2SF)	5	CRV	6330

Reaction No. 55672



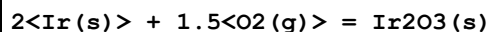
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-134.8(0.1SD)	5	ENS	5253

Reaction No. 55673



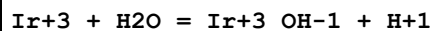
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.00(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:32.74(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:22.20(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:15.90(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:11.73(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-28.0(0.1SD)	0	ENS	5253
7	25	0	Inf. Dilution	dG:-45.03(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-58.0(3SF)	4	MTD	6422
9	27	0	Inf. Dilution	dH:-59.99(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-57.76(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-57.41(4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-57.02(4SF)	4	ENS	6422

Reaction No. 55674



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-42.0(0.1SD)	5	ENS	5253

Reaction No. 57428



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	1.05	Na(ClO4)/m	lgK:-4.78(3SF)	6	MHY	5271
2	15	1.05	Na(ClO4)/m	lgK:-4.59(3SF)	6	MHY	5271
3	25	1.05	Na(ClO4)/m	lgK:-4.37(3SF)	6	MHY	5271
4	35	1.05	Na(ClO4)/m	lgK:-4.13(3SF)	6	MHY	5271

Reaction No. 57429							
$\text{Ir+3_OH-1} + \text{H2O} = \text{Ir+3_OH-1(2)} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	1.05	Na(ClO4)/m	lgK:-5.63(3SF)	6	MHY	5271
2	15	1.05	Na(ClO4)/m	lgK:-5.27(3SF)	6	MHY	5271
3	25	1.05	Na(ClO4)/m	lgK:-5.20(3SF)	6	MHY	5271
4	35	1.05	Na(ClO4)/m	lgK:-5.08(3SF)	6	MHY	5271

Reaction No. 57430							
$\text{Ir+3_OH-1(3)_(s)} + 3\text{<H+1>} = \text{Ir+3} + 3\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	1.05	Na(ClO4)/m	lgK:11.10(4SF)	6	MHY	5271
2	15	1.05	Na(ClO4)/m	lgK:10.56(4SF)	6	MHY	5271
3	25	1.05	Na(ClO4)/m	lgK:10.22(4SF)	6	MHY	5271
4	35	1.05	Na(ClO4)/m	lgK:9.81(3SF)	6	MHY	5271

Reaction No. 57754							
$\text{Ir(s)} + 3\text{<Cl2(g)>} = \text{Ir+4_Cl-1(6)} - 2\text{<e-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-111.3(0.1SD)	0	ENS	5253

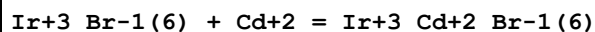
Reaction No. 57760							
$\text{Ir(s)} + 2\text{<O2(g)>} = \text{IrO4-2} - 2\text{<e-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-47.0(0.1SD)	4	ENS	5253

Reaction No. 57821							
$\text{Ba+2} + \text{Ir+3_Cl-1(6)} = \text{Ir+3_Ba+2_Cl-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.0(2SF)	1	ESC	
2	30	0.1	HClO4	lgK:-2.2(0.05SD)	5	MEF	5398
3	35	0.1	HClO4	lgK:-2.2(0.05SD)	5	MEF	5398
4	42	0.1	HClO4	lgK:-2.1(0.05SD)	5	MEF	5398

Reaction No. 57838							
$\text{Ir+3_Br-1(6)} + \text{Ba+2} = \text{Ir+3_Ba+2_Br-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

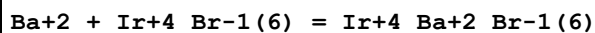
1	25	0.1	HClO4	lgK:2.8(0.04SD)	5	MEF	5398
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Reaction No. 57839



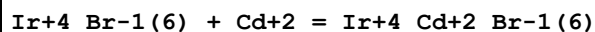
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	HClO4	lgK:2.9(0.1SD)	5	MEF	5398

Reaction No. 57840



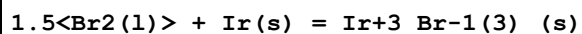
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	HClO4	lgK:2.3(0.1SD)	5	MEF	5398

Reaction No. 57841



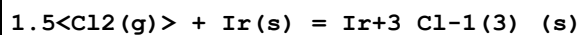
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	HClO4	lgK:1.6(0.1SD)	5	MEF	5398

Reaction No. 67359



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.04(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:23.85(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:14.91(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:9.163(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:5.377(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-32.79(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-42.50(4SF)	4	MTD	6422
8	27	0	Inf. Dilution	dH:-42.51(4SF)	4	MTD	6422
9	127	0	Inf. Dilution	dH:-52.93(4SF)	4	MTD	6422
10	227	0	Inf. Dilution	dH:-52.28(4SF)	4	MTD	6422
11	327	0	Inf. Dilution	dH:-51.60(4SF)	4	MTD	6422

Reaction No. 67360



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.69(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:29.43(4SF)	0	ENS	6422

3	127	0	Inf. Dilution	lgK:18.76 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:12.38 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:8.159 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-40.51 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-58.70 (4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-58.44 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-58.15 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-57.84 (4SF)	0	MTD	6422

Reaction No. 67361							
0.5<I2(s)> + Ir(s) = Ir+1_I-1_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.858 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:8.808 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:6.768 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:5.142 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:3.839 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-12.08 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-11.00 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-12.92 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-18.00 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-17.76 (4SF)	4	MTD	6422

Reaction No. 67362							
I2(s) + Ir(s) = IrI2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.04 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:14.95 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:11.24 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:8.204 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:5.741 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-20.52 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-20.0 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-23.85 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-34.03 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-33.55 (4SF)	4	MTD	6422

Reaction No. 67363							
$\text{Ir(s)} + 2\text{<S(s)>} = \text{Ir+4_S-2(2)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.71(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:21.57(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:15.75(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:12.11(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:9.635(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-29.62(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-31.80(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-32.96(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-33.71(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-34.25(4SF)	0	MTD	6422

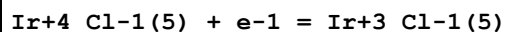
Reaction No. 67364							
$2\text{<Ir(s)>} + 3\text{<S(s)>} = \text{Ir+3(2)_S-2(3)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.40(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:34.18(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:24.98(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:19.25(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:15.35(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-46.93(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-50.20(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-51.98(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-53.13(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-53.96(4SF)	0	MTD	6422

Reaction No. 70259							
$2\text{<Ir+4_Cl-1(6)>} + \text{H}_2\text{O} = 2\text{<Ir+3_Cl-1(6)>} + 0.5\text{<O}_2\text{(g)>} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-7.15(3SF)	0	ENS	7276

Reaction No. 74296							
$\text{IrO(s)} + 2\text{<e-1>} + 2\text{<H+1>} = \text{Ir(s)} + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

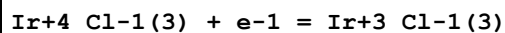
1	25	1	H2SO4	E:0.87(0.02SD)	4	CRV	22551
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Reaction No. 74297



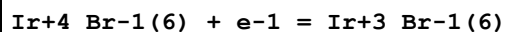
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.2	HNO3	E:1.0(2SF)	4	CRV	22551

Reaction No. 74298



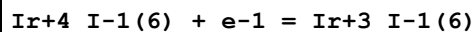
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.4	HClO4	E:1.30(3SF)	4	CRV	22551

Reaction No. 74299



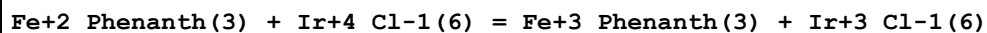
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	E:0.883(3SF)	4	CRV	22551

Reaction No. 74300



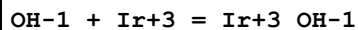
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KI	E:0.49(2SF)	4	CRV	22551

Reaction No. 74301



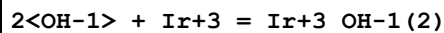
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	HCl	lgK:-2.46(0.06SD)	5	MEF	878

Reaction No. 77504



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.23(4SF)	1	ELC	

Reaction No. 77509



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.3(4SF)	1	ELC	

Reaction No. 77865

3<OH-1> + Ir+3 = Ir+3_OH-1(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.64(4SF)	1	ELC	

Reaction No. 77875							
4<OH-1> + Ir+4 + 2<Cl-1> = Ir+4_Cl-1(2)_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.13(4SF)	1	ELC	

Reaction No. 77877							
3<OH-1> + Ir+4 + 3<Cl-1> = Ir+4_Cl-1(3)_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:46.95(4SF)	1	ELC	

Reaction No. 77878							
2<OH-1> + Ir+4 + 4<Cl-1> = Ir+4_Cl-1(4)_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:40.91(4SF)	1	ELC	

Reaction No. 77879							
OH-1 + Ir+4 + 5<Cl-1> = Ir+4_Cl-1(5)_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.46(4SF)	1	ELC	

Reaction No. 79040							
2<S-2> + Ir+4 = Ir+4_S-2(2)_(s)							
Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1_S-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:133.2(4SF)	0	ELC	

Reaction No. 79041							
3<S-2> + 2<Ir+3> = Ir+3(2)_S-2(3)_(s)							
Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1_S-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:197.2(4SF)	0	ELC	

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EFE	Free Energy relations
EIE	Interpolation/Extrapolation (not Debye-Huekel)
ELC	Linear Combination of Reactions
ENS	Not specified
ESC	Standard conditions anchor
ESR	Similar reaction(s)
EVH	Van't Hoff Equation ($dH = \text{constant}$)
MCO	Colorimetry (often for measuring pH)
MDG	Combination of Thermodynamic Data
MEF	Electromotoric force, not specified
MHY	Emf with hydrogen electrode
MKN	Rate of reaction
MLC	Ligand competition
MPL	Polarography
MPT	Potentiometry
MSP	Spectroscopy

MTD	Temperature difference
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